

PHYSICS

Majoring & Minorin

The Physics Department offers majors and minors for the Bachelor of Science and the Bachelor of Arts degrees. Most students who obtain a BS in physics will go on to postgraduate studies in order to obtain an MS or Ph.D.

Students wishing to study **Engineering** are also supported by the department. For more information, please consult the [Applied Physics major](#) (Dual Degree, Pre-Engineering Program).

Major in Physics (BA): 8 courses

- PHY53-154 Fundamentals of Physics I
- PHY53-164 Fundamentals of Physics II
- PHY53-214 Modern Physics
- PHY53-324 Electromagnetism I
- PHY53-334 Classical Mechanics I
- PHY53-872 Capstone Course
- Two courses from:

PHY53-254 Observing the Universe

PHY53-364 Fundamentals of Materials Science and Engineering

PHY53-404 Electronics

PHY53-413 Classical Mechanics II

PHY53-424 Quantum Mechanics

PHY53-433 Electromagnetism II

PHY53-434 Statics

PHYS 473 Thermodynamics

Required supporting courses in the BA major: 3 courses

- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- MAT52-364 Modern Calculus III, or MAT52-754 Differential Equations

Major in Physics (BS): 10 courses

- PHY53-154 Fundamentals of Physics I
- PHY53-164 Fundamentals of Physics II
- PHY53-214 Modern Physics
- PHY53-324 Electromagnetism I
- PHY53-334 Classical Mechanics I
- PHY53-424 Quantum Mechanics
- PHY53-872 Capstone Course

- Three courses from:

PHY53-254 Observing the Universe

PHY53-364 Fundamentals of Materials Science and Engineering

PHY53-404 Electronics

PHY53-413 Classical Mechanics II

PHY53-433 Electromagnetism II

PHY53-434 Statics

PHY53-454 Math Methods in Physical Science

PHY53-463 Dynamics

PHY53-473 Thermodynamics

Required supporting courses in the BS major: 6 courses

- CHE311-103/101 General Chemistry I/Chemical Methods and Techniques Laboratory I
- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- MAT52-364 Modern Calculus III
- MAT52-754 Differential Equations

Note: Sufficient advanced mathematics for a mathematics minor is highly recommended.

Note: Each department at SU is required to provide a "capstone experience" for their majors. Past physics capstones have included experimental, observational and theoretical research in physics or a physics-related topic. Research has been carried out at SU, at off-campus REU programs and in industry. The detailed structure of the physics capstone experience is contractually agreed upon between the students and their capstone advisors. Generally, there are three components to the physics capstone: 1) the actual research work, 2) the writing of a significant paper on the research work and 3) an oral public presentation about the research work. Students seeking college credit for the research component of their capstone experience may (after consultation with their capstone advisor) sign up for Academic Internship (Physics 53-941, 942, 943, 944) or Independent Study and Research (Physics 53-951, 952, 953, 954). Regardless of whether the research phase of the capstone is done for credit the capstone course Physics 53-872 is required. It is a scientific writing course (see description below) that helps the student prepare both the capstone paper and oral presentation.

Minor in Physics: 5 courses

- PHY53-154 Fundamentals of Physics I
- PHY53-164 Fundamentals of Physics II
- PHY53-214 Modern Physics
- Two electives from:

PHY53-254 Observing the Universe

PHY53-324 Electromagnetism I

PHY53-334 Classical Mechanics I

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[PHY53-433 Electromagnetism II](#)

[PHY53-434 Statics](#)

[PHY53-454 Math Methods in Physical Science](#)

[PHY53-463 Dynamics](#)

[PHY53-473 Thermodynamics.](#)

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